

**Use of F test in comparison between two means:**

The F-test is not used to compare means directly — it is used to test whether two population variances are equal. This is crucial because the choice between the pooled t-test (which assumes equal variances) and Welch's t-test (which does not) depends entirely on the outcome of the F-test. Skipping this step risks using the wrong version of the t-test, which can lead to incorrect conclusions.

**Question#1 Comparing average tensile strength of steel rods from two suppliers**

A manufacturing firm is evaluating two steel suppliers — Supplier X (an established vendor) and Supplier Y (a new vendor with modern production facilities) — to determine which produces stronger steel rods. Quality engineers randomly selected 10 steel rods from each supplier and measured their tensile strength (in MPa). The recorded values are as follows:

Tensile Strength (MPa)

|            |     |     |     |     |     |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Supplier X | 412 | 428 | 435 | 419 | 408 | 431 | 416 | 440 | 424 | 430 |
| Supplier Y | 405 | 418 | 395 | 427 | 411 | 433 | 402 | 420 | 398 | 415 |

Test the hypothesis that the average tensile strength of steel rods from Supplier X is *no more than* that of Supplier Y. Use a significance level of  $\alpha = 0.05$ .

Solution : First we have to determine whether variance are equal or not:

Step#1  $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

Step#2 Level of significance:  $\alpha = 0.05$

Step#3 Test Statistics

$$F = \frac{s_1^2}{s_2^2}$$

Step#4

Critical region

$$F > F_{\frac{\alpha}{2}, v_1, v_2} \text{ or } F > F_{\frac{\alpha}{2}, v_2, v_1}$$

$$F > F_{0.025, 9, 9} \text{ or } F > F_{0.025, 9, 9}$$

$$F > 4.03 \text{ or } F > 4.03$$

If critical region satisfies reject  $H_0$

Step#5 Calculation:

|            |     |     |     |     |     |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Supplier X | 412 | 428 | 435 | 419 | 408 | 431 | 416 | 440 | 424 | 430 |
| Supplier Y | 405 | 418 | 395 | 427 | 411 | 433 | 402 | 420 | 398 | 415 |

$$s_1^2 = 107.3444, s_2^2 = 156.4889, n_1 = 10, n_2 = 10, \bar{X}_1 = 424.3, \bar{X}_2 = 412.4$$

$$F = \frac{s_1^2}{s_2^2} = \frac{107.3444}{156.4889} = 0.6860$$

0.6860 > 4.03 → doesn't satisfied

Step#6 Conclusion: Equal Variance

Now for mean:

Test the hypothesis that the average tensile strength of steel rods from Supplier X is *no more than* that of Supplier Y. Use a significance level of  $\alpha = 0.05$ .

Step#1  $H_0: \mu_1 \leq \mu_2$

$H_1: \mu_1 > \mu_2$

Step#2 Level of significance:  $\alpha = 0.05$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Step#4

Critical region

$$t > t_{\alpha, n_1 + n_2 - 2}$$

$$t > t_{0.05, 10 + 10 - 2}$$

$$t > t_{0.05, 18}$$

$t > 1.734$

If critical region satisfies reject  $H_0$

Step#5 Calculation:

$$s_1^2 = 107.3444, s_2^2 = 156.4889, n_1 = 10, n_2 = 10, \bar{X}_1 = 424.3, \bar{X}_2 = 412.4$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{(424.3 - 412.4) - (0)}{\sqrt{131.9167 \left( \frac{1}{10} + \frac{1}{10} \right)}} = 2.3168$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(9 * 107.3444) + (9 * 156.4889)}{18} = 131.9167$$

$2.3167 > 1.734 \rightarrow$  satisfied

Step#6 Conclusion: As  $t$  is satisfied therefore there is evidence to reject  $H_0$  at 0.05 level of significance we conclude that Average tensile strength of rod from supplier X is more than supplier Y.

Question#2 Comparing average crop yield between two fertilizer treatments

An agricultural research institute is studying the effect of two types of fertilizers on wheat crop yield. Fertilizer P is a conventional formula used for many years, while Fertilizer Q is a newly developed organic compound. Ten plots of equal size were randomly assigned to each fertilizer, and the yield (in kg per plot) was recorded at harvest time. The results are:

Crop Yield (kg per plot)

|              |     |     |     |     |     |     |     |     |     |     |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Fertilizer P | 182 | 196 | 175 | 203 | 188 | 191 | 179 | 207 | 194 | 185 |
| Fertilizer Q | 170 | 184 | 178 | 195 | 163 | 188 | 172 | 199 | 180 | 176 |

Test the hypothesis that the average yield from plots treated with Fertilizer P is *lesser than* that from plots treated with Fertilizer Q. Use a significance level of  $\alpha = 0.05$ .

Solution : First we have to determine whether variance are equal or not:

Step#1  $H_0: \sigma_1^2 = \sigma_2^2$

$H_1: \sigma_1^2 \neq \sigma_2^2$

Step#2 Level of significance:  $\alpha = 0.05$

Step#3 Test Statistics

$$F = \frac{s_1^2}{s_2^2}$$

Step#4

Critical region

$$F > F_{\frac{\alpha}{2}, v_1, v_2} \text{ or } F > F_{\frac{\alpha}{2}, v_2, v_1}$$

$$F > F_{0.025, 59, 59} \text{ or } F > F_{0.025, 59, 59}$$

If critical region satisfies reject  $H_0$

Step#5 Calculation:

|            |     |     |     |     |     |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Supplier X | 412 | 428 | 435 | 419 | 408 | 431 | 416 | 440 | 424 | 430 |
| Supplier Y | 405 | 418 | 395 | 427 | 411 | 433 | 402 | 420 | 398 | 415 |

$$s_1^2 =, s_2^2 =, n_1 =, n_2 =, \bar{X}_1 =, \bar{X}_2 =$$

$$F = \frac{s_1^2}{s_2^2} =$$

Step#6 Conclusion:

Now for mean:

Step#1  $H_0: \mu_1 - \mu_2$

$H_1: \mu_1 - \mu_2$

Step#2 Level of significance:  $\alpha=0.05$

Step#3 Test Statistics

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Step#4

Critical region

$$t > t_{\alpha, n_1 + n_2 - 2}$$

$$t > t_{0.05, 11 + 17 - 2}$$

$$t > t_{0.05, 26}$$

$$t > 1.706$$

If critical region satisfies reject  $H_0$

Step#5 Calculation: t

$$s_1^2 = , s_2^2 = , n_1 = , n_2 = , \bar{X}_1 = , \bar{X}_2 = ,$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Step#6 Conclusion:

## Chi-Square Test of Independence

The **Chi-Square Test of Independence** is a statistical method used to determine whether there is a significant relationship between two categorical variables.

It compares the observed frequencies in each category with the expected frequencies (if the variables were independent). If the difference is large, we conclude that the variables are likely related.

Identification Words: Relation, association, Dependency

Exercise: 11.2 Bluman Elementary Statistics

### Why we use it:

- To check if two variables are associated or independent
- Common in surveys, experiments, and social science research
- Example: Checking whether gender is related to product preference

## Contingency Table (Simple Definition)

A **contingency table** is a table that displays the frequency distribution of two or more categorical variables. It organizes data into rows and columns, making it easier to analyze the relationship between variables.

### Example:

|        | Like Product | Dislike Product |
|--------|--------------|-----------------|
| Male   | 20           | 10              |
| Female | 25           | 15              |

Steps of Applying Chi-Square Test of Independence:

Step 1

$H_0$ : The variables are independent

$H_1$ : The variables are dependent

Step 2 Level of Significance :  $\alpha$

Step 3: Test Statistics:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where E is define as

$$E = \frac{\text{Row total} * \text{Column total}}{\text{Grand total}}$$

Step 4:

$$\chi^2 > \chi^2_{\alpha, (r-1)*(c-1)}$$

Step 5: Calculation:

First calculate row total , Column total, then grand total, then make table

| Observed(O) | Expected(E) | $\frac{(O - E)^2}{E}$               |
|-------------|-------------|-------------------------------------|
|             |             |                                     |
|             |             |                                     |
|             |             |                                     |
|             |             | $\chi^2 = \sum \frac{(O - E)^2}{E}$ |

Step 6: Conclusion:

Question# **19. Choice of Exercise Equipment** Is the choice of exercise equipment dependent upon gender? Recent records from a large gym indicated the following equipment usage. At the 0.05 level of significance, is there a relationship?

|        | <b>Treadmill</b> | <b>Elliptical</b> | <b>Bike</b> |
|--------|------------------|-------------------|-------------|
| Male   | 120              | 60                | 75          |
| Female | 100              | 95                | 82          |

Step 1

$H_0$ : The variables are independent

$H_1$ : The variables are dependent

Step 2 Level of Significance :  $\alpha=0.05$

Step 3: Test Statistics:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where E is define as

$$E = \frac{\text{Row total} * \text{Column total}}{\text{Grand total}}$$

Step 4:

$$\chi^2 > \chi^2_{\alpha, (r-1)*(c-1)}$$

$$\chi^2 > \chi^2_{0.05, (2-1)*(3-1)}$$

$$\chi^2 > \chi^2_{0.05, 2}$$

$$\chi^2 > 5.991$$

If Cr satisfied reject Ho.

Step 5: Calculation:

First calculate row total , Column total, then grand total, then make table

| Gender       | Treadmill | Elliptical | Bike | Row total |
|--------------|-----------|------------|------|-----------|
| Male         | 120       | 60         | 75   | 255       |
| Female       | 100       | 95         | 82   | 277       |
| Column total | 220       | 155        | 157  | 532       |

| Observed(O) | Expected(E)          | $\frac{(O - E)^2}{E}$                      |
|-------------|----------------------|--------------------------------------------|
| 120         | (255*220)/532=105.45 | (120-105.45)^2/105.45=2.0076               |
| 60          | (255*155)/532=74.30  | (60-74.30)^2/74.30=2.7522                  |
| 75          | (255*157)/532=75.25  | (75-75.25)^2/75.25=0.0008                  |
| 100         | (277*220)/532=114.55 | (100-114.55)^2/114.55=1.8481               |
| 95          | (277*155)/532=80.70  | (95-80.70)^2/80.70=2.5340                  |
| 82          | (277*157)/532=81.75  | (82-81.75)^2/81.75=0.0008                  |
|             |                      | $\chi^2 = \sum \frac{(O-E)^2}{E} = 9.1435$ |

9.1435 > 5.991 → satisfied

Step 6: Conclusion: As CR satisfied therefore there is evidence to reject Ho at 0.05 level of significance we conclude Exercise methodology & gender are related to each other.